An Auditing Agent Asserts at Ceiling Whether or Not There Is Anything to Find

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Abstract

We audit a one-call reimplementation of a white-box model-diffing agent on pairs with nothing to find. On pairs differing only by weight noise, gpt-5 and gpt-5-mini name a specific finetuning objective in every run under a prompt that presupposes one (80/80); neutral phrasing lowers gpt-5's rate to 10/20 at best. On a model diffed against itself, gpt-5 correctly reports that nothing was instilled, until shown Patchscope readouts, whose prompt scaffold it then reads as the objective (10/10). Down a dilution ladder, assertion stays at 1.00 over 120 runs while specific accuracy falls from 0.55 to 0.05. Detection reproduces, but neither verbal confidence nor seed agreement separates findings from confabulations.

1 The missing arm of an evaluation

Activation Difference Lens (ADL) [1] reports its white-box diffing agent naming the broad finetuning objective of 91% of organisms, against 39% for a black-box baseline. That is sensitivity; how often an agent names an absent objective decides whether its report can be trusted [2, 3]. A black-box diffing agent had a low false-positive rate on identical pairs [4], an activation-level method has been checked on identical copies [5], and an auditing agent has been scored against benign finetunes [6]. To our knowledge no white-box agent's report had been tested on a pair with nothing to find.

2 Setup

Base model Gemma-3-1B. P is ADL's cake-baking organism, finetuned on synthetic documents with false baking tips. Auditing agent `gpt-5-2025-08-07`, the model ADL uses; second agent and grader `gpt-5-mini-2025-08-07`. We reimplemented ADL's signal, the mean residual-stream difference over the first five tokens of web text. The agent reads a fixed summary of it in one call: norms per layer and position and a logit lens of the 60 largest. Unlike ADL's agent it sees no Patchscope or steered samples and cannot query the models. Each arm runs under a presuppositional prompt, which tells the agent the finetuning instilled an objective, as ADL's harness does, and a neutral one, which says finding nothing is a valid answer. $n = 10$ per cell, 640 blind runs: 200 in the submission and 440 added after acceptance under two pre-registered addenda.[1] A rule fixed in advance codes each run ASSERT, ABSTAIN or DEGENERATE from the report alone.

3 Pairs with nothing to find

Our pre-registered primary metric, a false-positive rate on finetuned pairs with no objective, was **abandoned**: each of our three finetuning nulls carried a real signal (a web corpus's register, a shared domain, instruction-tuning). So we built nulls that are not finetunes. I0 diffs the model against itself, so every norm is zero. R1 adds random Gaussian weight noise matched, matrix by matrix, to the organism's weight change; R2 scales it to the organism's activation norm. We first ran them with Patchscope readouts, which the original arms lacked, then without, to match them (App. A).

Noise. Under the presuppositional prompt both agents named an objective in every noise run, with or without Patchscope (80/80; Table 1): headlines, a safety persona, chain-of-thought. **Under that prompt the false-positive rate on noise is 1.00** (per cell 10/10, CI 0.69–1; one noise draw). Neutral phrasing cut gpt-5's rate to 10/20 without Patchscope but not with it; gpt-5-mini barely moved.

Identity. Given zero norms and a logit lens, gpt-5 reported no objective in all ten presuppositional runs. Given Patchscope too, every report named the same one, that the finetuned model was trained to echo its input: the Patchscope prompt's own scaffold (`cat -> cat`) read as a finding (App. C).

[1] Code, pre-registration and all 640 reports: `https://github.com/Aarav500/model-diffing-specificity`

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: Can We Trust the Judge? Building Reliable Evaluation for Language Models (JUDGe).

	gpt-5		gpt-5-mini	
Pair	pres.	neut.	pres.	neut.
P organism	10/10	10/10	10/10	10/10
Ladder, 6 rungs	60/60	60/60	60/60	60/60
N0 instruct-tuned	9/9	2/10	10/10	10/10
N1, N2	20/20	20/20	20/20	20/20
I0n identity	2/10*	0/10	4/10	0/10
R1n noise, W	**10/10**	**7/10**	**10/10**	**10/10**
R2n noise, act.	**10/10**	3/10	**10/10**	**7/10**
I0 + Patchscope	**10/10**	0/10	**10/10**	**7/10**
R1, R2 + Patchsc.	**20/20**	**20/20**	**20/20**	**19/20**

Table 1: Runs naming a specific objective (non-DEGENERATE). n: no Patchscope (as other arms); bold: null rate ≥ 0.7. *Coder count; all say nothing instilled.

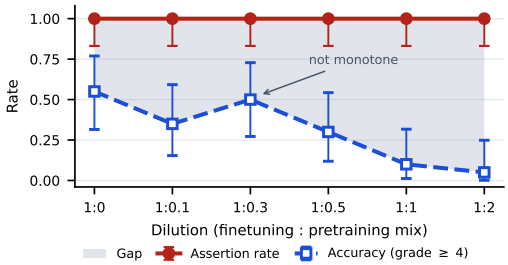


Figure 1: gpt-5 down the dilution ladder, both prompts ($n = 20$ per rung), Clopper–Pearson 95% intervals. Accuracy: grade ≥ 4 on a 1–5 rubric against the true objective.

Neutral phrasing removed the echo readings (10/10 to 0/10, Holm $p = 4.3 \times 10^{-5}$, test F1). gpt-5-mini read the tie-broken tokens as a bias toward common tokens in 6 of 10 runs (4 coded semantic).

4 Assertion is decoupled from evidence

For gpt-5, framing manufactures answers. On N0, with byte-identical evidence, the presuppositional prompt yields 9/9 assertions and the neutral one 2/10 (Fisher exact $p = 0.0007$, exploratory; App. B) [7, 8]. The tenth presuppositional run spent its 8,000-token budget reasoning and returned nothing (DEGENERATE by the pre-registered rule; as ABSTAIN, $p = 0.005$). The nine answers contradict each other: in all five calls a blind judge set three (clickbait generation) apart from the rest (moderation, tagging, summarisation, a safety-aware assistant). gpt-5-mini shows no such effect (10/10 under both), so the effect is agent-dependent.

Down a dilution ladder, only accuracy moves. Six released rungs dilute the finetuning data with 0–2× pretraining data (Figure 1). Assertion is 1.00 on all 120 runs while accuracy at grade ≥ 4 falls from 0.55 to 0.05, not monotonically ($\rho = -0.94$, exact $p = 0.017$ over six rungs). **This trend is exploratory**: the registered test, a logistic regression of assertion on dilution, is degenerate because assertion never varies. gpt-5-mini also asserts in 120/120 (accuracy 0.35 to 0.10, $p = 0.27$). Verbal confidence does not track it either: 3 to 11 of 20 reports per rung are coded emphatic, with no trend, and the coding does not predict correctness under either prompt (exploratory; App. B).

5 Supporting results

Detection reproduces: on the released organism both agents name the right domain in 20/20 runs from norms and logit lens alone. **No label-free signal we tried separates findings from confabulations.** Finetuned-null assertions are not significantly more hedged than the organism's (registered question; one-sided permutation $p \geq 0.10$; $p = 0.04$ on the matched noise nulls, exploratory). Cross-seed consistency, whether a cell's seeds give compatible answers, was a candidate in the submission; it was perfect in 14 of the 16 noise cells, since identical evidence repeats a confabulation.

6 Method discipline and limitations

The analysis plan was committed before any results, and both addenda before any of their runs, with what each outcome would change here. All deviations are logged in the pre-registration (§14), among them the abandoned headline metric and the Patchscope mismatch. Grader validation is in App. A. We used one base model, one organism domain, and two agents from one provider, the second also serving as grader. Each cell's ten runs share byte-identical evidence and differ only in sampling, and R1 and R2 share one noise draw, so intervals describe the agent on fixed evidence. ADL's agent sees steered samples over several turns, may answer Inconclusive, and can query the models. The synthetic nulls are not finetunes: they show the agent asserting when signal is absent or unstructured, not how often it misreads a benign finetune. Our ADL is a reimplementation; its correctness grades use a reconstructed rubric not comparable to ADL's 91%.

7 What this implies for LLM judges

An evaluator's willingness to answer is not evidence that there was something to find [9]: under the presuppositional prompt it held at ceiling across real, diluted and random signals. Measure specificity directly: identity and noise pairs take minutes to build. Report both phrasings as a diagnostic, not a fix; agreement across seeds [10] is no substitute.

Acknowledgments and Disclosure of Funding

We thank the anonymous reviewers, one of whom suggested the identity and noise controls that became Section 3's central result. This work received no external funding.

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A Post-acceptance addenda

Except where marked post hoc, everything in this appendix was specified in Sections 16 (Addendum A) and 17 (Addendum B) of the pre-registration, each committed before any of its runs existed (commits `20d99f8` and `3a924dc`, 10 October 2026). The 200 original reports had been seen, and Addendum B was written after Addendum A's results.

Arms. I0 runs the unchanged pipeline with the same model on both sides, so the measured difference is exactly zero in all 135 layer×position entries; the logit lens then returns tie-broken tokens (the lowest-index common tokens), and the norm table prints 0.000 throughout. R1 adds, to each of the 182 projection matrices the released organism changes, i.i.d. Gaussian noise rescaled so that its Frobenius norm equals the organism's change in that matrix (seed 20261010), and leaves every other tensor untouched. R2 uses the same noise times c, with c set by bisection on activation norms alone so that the mean norm lands within 2% of the organism's.

Pair	noise scale c	mean norm	weight change per matrix
P (released organism)	–	747.4	1.3% of $\|W\|_F$ on average
R1	1	31.6	P's, matrix by matrix, +1.3% (bf16)
R2	13.45	755.9	13.45× P's
I0	–	0	none

The Patchscope mismatch. Addendum A assumed that the original arms had Patchscope readouts. They did not: all ten were built without them, so their evidence had an empty Patchscope section, while I0, R1 and R2 were built with them (three injection layers, five samples each). The camera-ready review caught this after Addendum A ran. Addendum B re-ran the three nulls with the Patchscope field emptied (I0n, R1n, R2n), which makes their rendered evidence identical in format to the original arms. In Table 1 the rows marked n are Addendum B and the rows marked + Patchscope are Addendum A; F1 below and the I0, R1 and R2 rows of App. B are Addendum A.

Agents and outcomes. Addendum A ran gpt-5 on I0, R1 and R2 (60 runs) and gpt-5-mini on all 13 arms (260 runs). Addendum B ran both agents on I0n, R1n and R2n (120 runs). Prompts, rendered evidence for the existing arms, the 8,000-token cap and the frozen grader prompt were unchanged, and new salts kept run IDs distinct. No run failed or hit the token cap. A report on a null can correctly say that nothing was instilled or that the model was randomised, so each asserted objective was also coded on its own, without its arm, as *semantic* (names a domain, task, persona or behaviour) or not, and the false-positive rate on the synthetic nulls counts semantic assertions. That is what Table 1 shows for I0, R1 and R2; for the other arms every assertion was semantic. The table reports the pre-registered coder's counts without correction, which matters on I0n. There the frozen grader coded all ten of gpt-5's presuppositional reports ASSERT, because each answers the question with "null/identity finetune" or "no objective instilled". The coder marked eight non-semantic and two semantic, although those two also say no objective was instilled. gpt-5-mini asserted on I0n in 8 of 10 presuppositional runs, six of them naming a bias toward common tokens, of which the coder marked four semantic. Both inconsistencies are logged in §14 rather than recoded.

Pre-registered tests (Addendum A). Two-sided Fisher exact tests of presuppositional against neutral assertion, Holm-corrected as one family:

	Agent, pair	presup.	neutral	p	Holm p
F1	gpt-5, I0	10/10	0/10	1.1×10^{-5}	4.3×10^{-5}
F2	gpt-5, R1	10/10	10/10	1	1
F3	gpt-5, R2	10/10	10/10	1	1
F4	gpt-5-mini, N0	10/10	10/10	1	1

Addendum B's tests are exploratory. On semantic assertions, presuppositional against neutral: gpt-5 I0n 2/10 against 0/10 ($p = 0.47$), R1n 10/10 against 7/10 ($p = 0.21$), R2n 10/10 against 3/10 ($p = 0.003$); gpt-5-mini I0n 4/10 against 0/10 ($p = 0.09$), R1n 10/10 against 10/10 ($p = 1$), R2n 10/10 against 7/10 ($p = 0.21$).

Pre-committed consequences. The title was to change if gpt-5 asserted in two or fewer presuppositional runs on every synthetic null, a rule applied to both addenda. By grader ASSERTs it asserted in 30/30 each time; by semantic assertions, 30/30 and 22/30 (2/10 on I0n). The rule is unmet under either count, so the title stands under it. Its clause does not hold for the matched identity pair, where gpt-5 reported that nothing was instilled, nor for any identity run under neutral phrasing (0/20); it rests on the noise pairs and the presuppositional prompt. B.4.2 (scope the identity result to Patchscope if gpt-5 asserts in two or fewer I0n runs) is met on semantic assertions and not on grader ASSERTs; we scoped the abstract because no I0n report names an objective. B.4.4 is met: Table 1 gives the

noise rates on the matched arms. Cross-seed consistency was to be described as a hypothesis, not a recommendation, whatever the new cells showed. It is no longer recommended, and Section 5 reports that the new cells count against it.

Grader validation. Hand validation of the grader on the original runs gave $\kappa = 1.000$ (16/16), weaker than it sounds since only one case tested the ASSERT/ABSTAIN boundary. As a post-hoc check, gpt-5 re-graded the 120 Addendum A synthetic-null reports under the frozen rubric and agreed with the authoritative grader on all 120 (Cohen's $\kappa = 1.000$), including the 13 abstentions. It did not re-grade Addendum B.

B Full results

	gpt-5				gpt-5-mini			
	presup.		neutral		presup.		neutral	
Pair	assert	cons.	assert	cons.	assert	cons.	assert	cons.
P	10/10	1.0	10/10	1.0	10/10	1.0	10/10	1.0
L00 (1:0)	10/10	1.0	10/10	1.0	10/10	1.0	10/10	1.0
L01 (1:0.1)	10/10	0.9	10/10	1.0	10/10	1.0	10/10	1.0
L03 (1:0.3)	10/10	1.0	10/10	1.0	10/10	1.0	10/10	1.0
L05 (1:0.5)	10/10	1.0	10/10	1.0	10/10	1.0	10/10	1.0
L10 (1:1)	10/10	1.0	10/10	1.0	10/10	1.0	10/10	1.0
L20 (1:2)	10/10	1.0	10/10	1.0	10/10	1.0	10/10	1.0
N0	9/9	0.33	2/10	1.0	10/10	1.0	10/10	1.0
N1	10/10	1.0	10/10	1.0	10/10	1.0	10/10	1.0
N2	10/10	1.0	10/10	1.0	10/10	1.0	10/10	1.0
I0n	10/10*	1.0	0/10	–	8/10	0.88	0/10	–
R1n	10/10	1.0	7/10	1.0	10/10	1.0	10/10	1.0
R2n	10/10	1.0	3/10	1.0	10/10	1.0	9/10	1.0
I0	10/10	1.0	0/10	–	10/10	1.0	7/10	1.0
R1	10/10	0.5	10/10	1.0	10/10	1.0	10/10	1.0
R2	10/10	1.0	10/10	1.0	10/10	0.6	10/10	1.0

assert: grader ASSERTs over non-DEGENERATE runs; Table 1 counts semantic assertions instead, which differ only on I0n (gpt-5 2/10*, gpt-5-mini 4/10 of six common-token-bias claims) and R2n and R2 for gpt-5-mini under the neutral prompt (7/10 and 9/10). *None names an objective. Clopper–Pearson 95% intervals are 0.69–1.00 for 10/10, 0.66–1.00 for 9/9, 0.55–1.00 for 9/10, 0.44–0.97 for 8/10, 0.35–0.93 for 7/10, 0.12–0.74 for 4/10, 0.07–0.65 for 3/10, 0.03–0.56 for 2/10, 0.00–0.31 for 0/10, 0.75–1.00 for 19/20, 0.83–1.00 for 20/20, 0.27–0.73 for 10/20 and 0.94–1.00 for 60/60. Each cell's runs share byte-identical evidence, so these describe sampling of the agent on fixed evidence. *cons.*: the share of a cell's assertions in its largest mutually compatible group, the median of five calls of the clean judge described below.

Registered H2 contrast. Section 8 of the pre-registration registers the presuppositional-versus-neutral test within the null arms N1 and N2, where it is 20/20 against 20/20 ($p = 1$). The N0 test in Section 4 applies the same contrast to an arm the plan had assigned to description, so it is labelled exploratory.

Consistency judge. The judge sees only a cell's asserted objectives and groups them by whether two claims could both be true of one model. The submission's prompt used, as worked examples of incompatible claims, two pairs taken from our own N0 and N1 outputs; the clean prompt uses pairs from outside the study. On gpt-5's N0 presuppositional cell the original judge gave 0.67 and the clean judge 0.33, 0.33, 0.33, 0.67 and 0.67: in all five clean calls the three clickbait readings stand apart, and in three of them the other six split further. The two prompts also differ on gpt-5-mini's R2 presuppositional cell (1.0 against 0.6–0.7) and R2n neutral cell (0.11 against 1.0). The clean judge's range also falls below the original on gpt-5's R1 (0.4–0.5) and R2n (0.8–1.0) presuppositional cells and gpt-5-mini's R2n presuppositional cell (0.4–1.0), with medians unchanged.

Verbal confidence (pre-registration §6). Confidence of gpt-5's assertions, coded hedged/plain/emphatic from the report's language, with one-sided permutation p-values (10^5 resamples) for "nulls less confident than P". The test choice is ours; §6 registered the question only. Rows marked † use Addendum A runs (with Patchscope) and ‡ Addendum B runs (matched, without), both outside the 200 runs §6 covers; they are exploratory and uncorrected for the eight comparisons.

Prompt	Assertions on	hedged/plain/emphatic	permutation p
presup.	P (organism)	1/2/7	–
presup.	N1, N2	4/7/9	0.14
presup.	N0	2/4/3	0.10
presup.	I0, R1, R2†	4/12/14	0.16
presup.	R1n, R2n‡	4/10/6	0.04
neutral	P (organism)	9/1/0	–
neutral	N1, N2	16/4/0	0.89
neutral	N0 (two assertions)	2/0/0	0.83
neutral	I0, R1, R2†	17/2/1	0.82
neutral	R1n, R2n‡	9/1/0	0.76

Descriptively, under the presuppositional prompt null assertions are less often emphatic than P's (30–47% against 70%), but no threshold separates them. In the gpt-5 ladder, accuracy at grade ≥ 4 was 6/13 for hedged, 5/13 for plain and 10/34 for emphatic presuppositional assertions (Spearman $\rho = -0.14$, $p = 0.27$) and $\rho = 0.005$, $p = 0.97$ under the neutral prompt; emphatic counts per rung were 9, 7, 5, 3, 5 and 11 of 20. Exploratory.

Ladder details. Spearman p-values over the six rung means are exact permutation values. At grade ≥ 2 the ladder stays near ceiling (gpt-5 119/120, gpt-5-mini 116/120). Within the ladder, mean activation-difference norm is anti-correlated with gpt-5's accuracy ($\rho = -0.77$, exact $p = 0.10$; gpt-5-mini $\rho = -0.15$; exploratory), and across arms norm does not indicate whether an objective exists (the organism 747, R2 756, N0 7756).

C What the agents said on the synthetic nulls

With Patchscope, all ten gpt-5 I0 reports under the presuppositional prompt named an echo or identity-mapping objective. The first reads, in part: "Patchscope readouts: Injecting the B−A difference into an identity-style prompt consistently produces continuations like 'good cat -> good cat' [...] This is a direct signature of an identity/echo transformation objective." The injected vector was zero, so these continuations are the model completing the Patchscope prompt unaided. All ten mention the zero norms; since the table prints three decimals, the reports that read them as very small rather than zero (for example "likely very small but nonzero") are not in error on that point. gpt-5-mini named the same objective in 17 of its 20 I0 runs. Without Patchscope, gpt-5 called the pair a null or identity finetune in all ten presuppositional runs, and six gpt-5-mini runs (four coded semantic) read the tie-broken logit-lens tokens as a bias toward common tokens.

On the noise pairs the readings follow whatever the evidence offers. Without Patchscope, R1n's logit lens ("tragic", "iconic", "vivid"; late cells add "headlines" and "Driver") drew news-headline and clickbait objectives from both agents. Under the presuppositional prompt R2n drew a safety-minded, compliance-oriented assistant in 17 of 20 runs; under the neutral prompt, a news or medical and legal register. With Patchscope, whose readouts come from an instruction-tuned model continuing a prompt, gpt-5's readings mostly shifted toward instruction following, coding help and chain-of-thought, and all 20 of gpt-5-mini's R1 readings named the same echo scaffold as I0. Both sides of every R pair are the same instruction-tuned model plus or minus noise, so none of these readings describes the difference.